Quadratics: Expanding & Factorising

Practice test 8 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 3x + 10x + 30$ (1) = $\mathbf{x^2 + 13x + 30}$ (1)

A2. $x^2 + 13x – 4x – 52$ (1) = $\mathbf{x^2 + 9x – 52}$ (1)

A3. $x^2 + 5xy – 2xy – 10y^2$ (1) = $\mathbf{x^2 + 3xy – 10y^2}$ (1)

A4. $(3x^2 + 12x + 12) – (x^2 + 3x – 4)$ (1) = $\mathbf{2x^2 + 9x + 16}$ (1)

Section B — Expanding three brackets

B1. $(x – 4)(x + 1) = x^2 – 3x – 4$ (1) $(x^2 – 3x – 4)(x + 3) = x^3 + 3x^2 – 3x^2 – 9x – 4x – 12$ (1) = $\mathbf{x^3 – 13x – 12}$ (1)

B2. $(2x + 1)^2 = 4x^2 + 4x + 1$ (1) $(4x^2 + 4x + 1)(2x + 1) = 8x^3 + 4x^2 + 8x^2 + 4x + 2x + 1$ (1) = $\mathbf{8x^3 + 12x^2 + 6x + 1}$ (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to 21, add to 10: 3 and 7 (1) **(x + 3)(x + 7)** (1)

C2. Multiply to –36, add to –9: –12 and 3 (1) **(x – 12)(x + 3)** (1)

C3. $(5x)^2 – 8^2$ = **(5x + 8)(5x – 8)** (1)

C4. $2x(x^2 – 9)$ (1) = **2x(x + 3)(x – 3)** (1)

Section D — Factorising $ax^2 + bx + c$

D1. $2 \times –20 = –40$; numbers: 8 and –5, so $2x^2 + 8x – 5x – 20 = 2x(x + 4) – 5(x + 4)$ (1) **(2x – 5)(x + 4)** (1)

D2. $3 \times 10 = 30$; numbers: –15 and –2, so $3x^2 – 15x – 2x + 10 = 3x(x – 5) – 2(x – 5)$ (1) **(3x – 2)(x – 5)** (1)

D3. $6 \times –5 = –30$; numbers: –15 and 2, so $6x^2 – 15x + 2x – 5 = 3x(2x – 5) + 1(2x – 5)$ (1) **(3x + 1)(2x – 5)** (1)

Section E — Using factorising

E1. $x^2 + 4x – 21 = 0$ (1) $(x + 7)(x – 3) = 0$ (1) **x = –7 or x = 3** (1)

E2. Numerator = $3(x + 2)(x – 2)$ (1) denominator = $(x + 3)(x – 2)$ (1) cancel $(x – 2)$: **3(x + 2) / (x + 3)** (1)

Section F — Problem solving & proof

F1. $3(x^2 – 4x) + 5$ (1) = $3[(x – 2)^2 – 4] + 5$ (1) = $\mathbf{3(x – 2)^2 – 7}$ (1)

F2. The outer square has sides of $(x + 4)$ m (1) $(x + 4)^2 = x^2 + 8x + 16$ (1) path = $(x^2 + 8x + 16) – x^2$ = **(8x + 16) m^2** (1)

F3. (12.5 + 7.5)(12.5 – 7.5) = 20 × 5 = **100** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: (x – 7)(x + 4) is the same as (x + 4)(x – 7). “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.